
Language Models Reproduce Reasoning Procedures but Not Answers Under Verified Contamination

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Abstract

A benchmark item found in a model’s training data is routinely taken to mean its score on that item reflects recall, but testing the inference requires knowing what the model read, which the corpora of deployed systems do not permit. Instella publishes its full pretraining mixture and checkpoints from every stage, and its stage two corpus derives from the GSM8K training split, so membership is fixed by exact thirteen gram containment rather than estimated: against the 1319 test items, known negatives because the corpus derives only from train, the adopted threshold flags none, a Wilson upper bound of 0.29 percent. A premise deletion probe over 1591 train and 1588 test variants, across 891 and 888 parent problems, finds no rise in answer reproduction where the corpus enters training: a difference in differences of -0.26 points on $[-1.66, +1.14]$, holding at $+0.25$ and $+0.94$ across the later boundaries, and unchanged under restriction to small answers, to terminated generations, and to a five bin containment gradient within the train arm that runs 3.7, 1.5, 0.9, 4.3 and 4.1 percent with no monotone order, against a design resolution of 1.89 points at 80 percent power. Continued pretraining at five doses under a fixed token budget explains the null and separates two quantities usually conflated: at 64 repetitions the model reproduces 94.6 percent of injected text verbatim, yet against a never injected control the memorized solution procedure returns at $+4.76$ points $[+1.30, +8.23]$ while the final answer does not, at $+1.52$ points $[-1.93, +5.08]$, procedure rising monotonically with dose and answer flat throughout. Exposure is legible in the calculation steps and not the answer, and needs far more repetition than the corpus supplies to any item, where document multiplicity reaches four. Two further checks locate what does cost accuracy: appending a dependent step to problems the strongest checkpoint already solves leaves 88.0 percent correct with zero reversions in 418 trials, while a crossed design holding sentence length, syntax and position fixed finds the main effect of an inserted unused number to be null at every checkpoint, -0.59 $[-1.62, +0.45]$, -0.79 $[-1.93, +0.31]$ and -0.42 $[-1.53, +0.68]$, with topicality null too and a residual cost of about two points for any insertion at all. Two perturbation designs indicating heavy fragility, an externally authored clause ladder and premise reordering, did not survive verification and are reported as withdrawn. Intervals are cluster bootstraps over parent problems, 4000 draws.

1 Introduction

Finding a benchmark item in a model’s training data is, by convention, grounds for discounting the score on that item. The step from the one to the other is rarely tested, since a test needs both certainty about what the model read and access to it at more than one point in training. Neither is available for frontier systems. Detection has grown up as a literature of proxies instead, from n gram overlap [Brown et al., 2020] to perplexity based membership inference [Carlini et al., 2023, Shi et al., 2024], and plain rephrasing is enough to defeat the overlap rules [Yang et al., 2023]. All

40 of these estimate membership. None establishes it, and an estimated treatment label cannot tell a
41 real null apart from an uninformative one.

42 Instella [Liu et al., 2025] is the exception this work relies on. AMD publishes the complete pretrain-
43 ing mixture together with checkpoints from every stage, including the stage two synthetic mathemat-
44 ics corpus built from the GSM8K training split [Cobbe et al., 2021]. Most released models expose
45 only final weights; Instella exposes the text consumed at each stage, so whether a given problem
46 entered training, and when, becomes a matter of direct containment. Every GSM8K test item is
47 a known negative with respect to that corpus, which turns detector error into a measured quantity:
48 at the threshold adopted here, zero of 1319 test items are flagged, a Wilson upper bound of 0.29
49 percent.

50 Knowing an item was available to be memorized says nothing about whether memorizing it inflated
51 the score, so the probe removes one premise the solution genuinely uses. The original answer is then
52 no longer derivable, and a model that remembered the parent problem can still emit it while a model
53 working from the remaining text cannot. Instella ships four checkpoints along one trajectory, stage
54 one before the synthetic corpus appears, stage two where it enters, an instruction tuned checkpoint
55 and a preference optimized one, so the same items go to the model on both sides of that boundary.
56 A memorization account predicts the rate climbs once the corpus arrives. It does not. To find what
57 the probe could have detected, selected items were injected by continued pretraining at controlled
58 doses, and there verbatim reproduction and answer recall come apart sharply.

59 2 Setup

60 **Corpus and checkpoints.** The stage two corpus, amd/Instella-GSM8K-synthetic, ships a
61 generated pool and a 119 thousand item subset, train_119K, documented as the split actually con-
62 sumed in stage two pretraining. Containment is measured against that subset as a maximum over
63 individual documents, so a claim of exposure points at one training document and not at the corpus
64 treated as a single pooled text. Four checkpoints are probed: Stage 1, which precedes the corpus; In-
65 stella 3B, where it enters; SFT; and Instruct. Generation runs on vLLM [Kwon et al., 2023], greedy
66 for the primary estimates, with a temperature sweep as a check.

67 **Premise deletion.** One sentence the calculator annotated solution consumes is dropped from each
68 GSM8K train and test parent problem. Two confounds correlated with containment are handled at
69 construction. A necessity check confirms the deleted value appears nowhere else in the prompt, and
70 variants whose original answer survives elsewhere are thrown out, since emitting it there is copying,
71 and that leakage runs about twice as often on high containment items. What remains is 1591 train
72 and 1588 test rows over 891 and 888 parents. Hit rate counts variants where the model emits the
73 deleted parent’s original answer.

74 **Controlled injection.** Calibration comes from injecting 200 verified clean GSM8K test parents,
75 alongside 200 held out entirely, into Instella 3B by continued pretraining. The token budget is fixed
76 across five doses of zero, one, four, sixteen and sixty four repetitions, so a higher dose displaces
77 generic filler and does not buy extra compute. Injected document loss and verbatim continuation sit
78 inside every arm as positive controls, since a silently failed run and a genuine absence of memoriza-
79 tion both give a flat curve. The same probe runs on both sets at every dose.

80 **Composition and perturbation.** A depth extension probe appends one dependent calculation step
81 to problems the model already gets right, changing the gold answer, and asks whether the step
82 is composed or the model falls back on the original problem. Answer preserving perturbations,
83 paraphrase, an inserted off topic sentence, entity substitution, and a sentence carrying one unused
84 number, are each scored against the same problem’s own original.

85 3 Results

86 3.1 Recall across the training trajectory

87 Figure 1 gives the hit rate on GSM8K train, which sits in the stage two corpus, and on GSM8K test,
88 absent from every corpus scanned, together with the difference in differences at each boundary. A

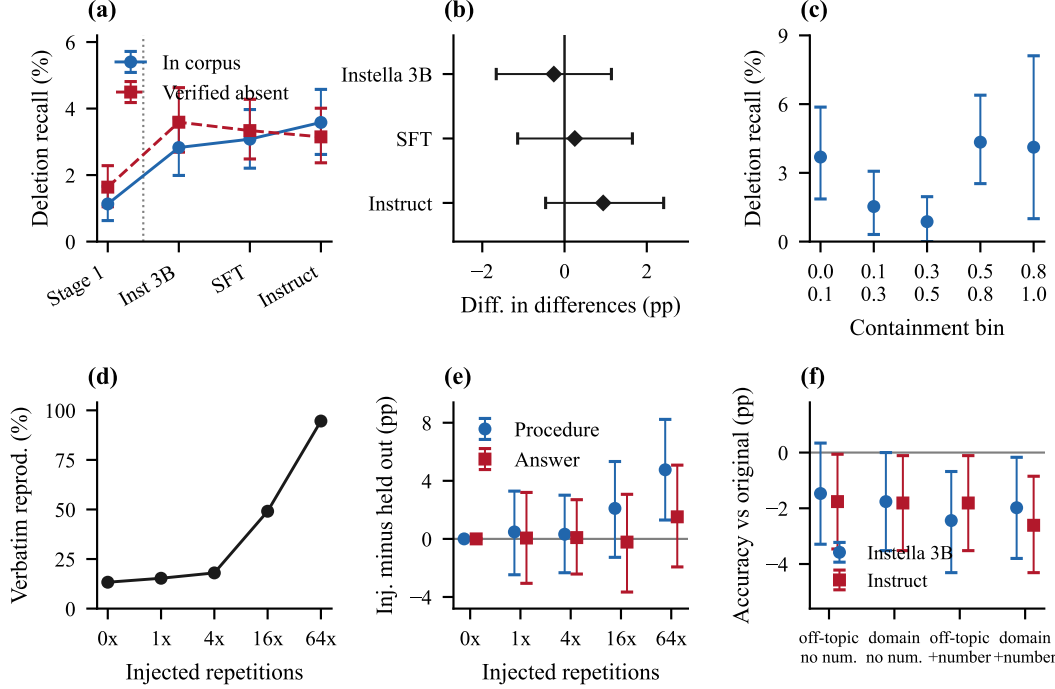


Figure 1: (a) Premise deletion recall by checkpoint for items inside the stage two corpus and for verified clean items; the dotted line marks where the corpus enters. (b) Difference in differences between the arms at each training boundary. (c) Recall within the train arm alone, binned by containment against the consumed corpus, which uses no train against test contrast. (d) Verbatim reproduction of injected documents against dose, confirming the intervention took effect. (e) Reproduction of the memorized solution procedure and of the final answer, each against the never injected control. (f) Crossed design isolating the inserted number: four twenty word insertions differing only in an off topic against a problem derived noun and a filler word against an unused integer, each against its own untouched problem. Error bars are 95 percent cluster bootstraps over parent problems, 4000 draws.

89 memorization account puts the gap between stage one and stage two. Per checkpoint the train arm
90 runs 1.13, 2.83, 3.08 and 3.58 percent against 1.64, 3.59, 3.34 and 3.15 percent on the test arm, over
91 1591 and 1588 rows.

92 No gap opens at the boundary, and what movement there is runs the wrong way. The difference
93 in differences comes to -0.26 points on an interval of $[-1.66, +1.14]$, and it stays flat at every
94 later boundary too, $+0.25$ into SFT and $+0.94$ into Instruct. Three variations leave it where it is:
95 restricting to one and two digit answers, restricting to rows where both checkpoints terminated, and
96 replacing the train against test contrast with a five bin containment gradient inside the train arm
97 alone. That last one matters most, since it drops the two arm comparison entirely, and the bins run
98 3.7, 1.5, 0.9, 4.3 and 4.1 percent from least to most contained, with no order to them. Over 891
99 parent clusters the design resolves 1.89 points at 80 percent power. Every gap observed falls under
100 that.

101 3.2 Procedure and answer under controlled injection

102 Injection worked: verbatim reproduction climbs from 13.3 percent with no injection to 94.6 percent
103 at the top dose. Figure 1 sets two outcomes against the never injected control at each dose, whether
104 the deleted premise value comes back and whether the memorized intermediate calculation steps
105 come back. Procedure reproduction runs $+0.48$, $+0.32$, $+2.10$ and $+4.76$ points across the four
106 nonzero doses, the last on $[+1.30, +8.23]$; answer reproduction runs $+0.06$, $+0.09$, -0.22 and
107 $+1.52$, the last on $[-1.93, +5.08]$.

108 Answer recall never separates from the control, even where the model has all but memorized the
109 source text. Procedure reproduction climbs steadily and clears zero at the top dose. So the model

can reproduce a document almost word for word and still not recover the quantity that document deletes, or the answer that quantity fixes. Exposure shows up in the calculation steps. It takes far more of it than the released corpus ever supplies to one item, where document repetition tops out at four.

3.3 Composition and surface sensitivity

If remembering is not the constraint, two candidates remain: the model cannot chain an extra step, or any change to a problem’s surface throws it. Neither holds. Given a problem Instruct already solves and one dependent step appended, it stays correct on 88.0 percent, 368 of 418, and never falls back on the pre extension answer. Panel (f) of Figure 1 takes the second candidate apart.

An earlier comparison suggested the inserted number was what mattered, since a sentence carrying an unused quantity cost 6.2 points where a plain distractor sentence cost 0.6. Those conditions also differed in length and topical relatedness, so a crossed design separates them: all four insertions are the same twenty word frame with two words swapped, one choosing an off topic or a problem derived noun and one a filler word or a free two digit integer, so length, syntax and position are identical across cells. The main effect of the number was fixed as the primary endpoint before the run, and it is null at every checkpoint, -0.59 on $[-1.62, +0.45]$ at stage two, -0.79 on $[-1.93, +0.31]$ after fine tuning and -0.42 on $[-1.53, +0.68]$ at Instruct, over 1764 parents each. Topicality is null too. What survives is the cost of insertion itself, in panel (f) of Figure 1: all four cells sit near -2 points with overlapping intervals. The earlier 6.2 point figure was carried by length and topicality, not by the quantity.

3.4 Two withdrawn checks

Two designs looked at first like heavy evidence of fragility, and neither held up. A perturbation ladder written elsewhere seemed to drop accuracy by over 30 points under two added clauses; its later rungs are independently generated problems, harder and structurally different, not the same items with clauses bolted on, so part of the drop is difficulty. Ivanova [Ivanova, 2024] makes the general form of the point, that an estimated difference wants an interval before an interpretation. The second was premise reordering, apparently worth over 12 points at every checkpoint, whose generator permutes premises with no check on whether a later sentence depends on an earlier one, leaving many variants simply broken. Restricted to items free of such dependency, 24 percent of the set, the effect lands on $+2.2$ points over $[-6.5, +10.9]$ across 46 problems. Neither supports anything here.

4 The unavailable contrast

The obvious design contrasts GSM8K train items at high containment against those at low, and it was tried first. Two findings closed it off. The subset the model actually consumed, `train_119K`, puts 7.56 percent of train items at containment 0.80 or above against 29.75 percent computed on the full released pool, so a pool defined arm draws about half its items from text the model never read. The second is decisive: `allenai/tulu-3-sft-mixture` [Lambert et al., 2024] sits in the stage two mixture and `nvidia/OpenMathInstruct-2` [Toshniwal et al., 2024] arrives at fine tuning, each carrying GSM8K train at containment 0.999 or above on over 97 percent of items against under 0.1 percent for test. Every train item therefore reached the model after stage one by some route, so a high against low split on the Instella corpus alone compares two saturated groups. Appendix B gives the per corpus figures. The checkpoint design never sorts items by containment and escapes this entirely.

5 Limitations and conclusion

One family, one scale near three billion parameters, one benchmark domain, and a dissociation shown at five doses rather than ruled out at any exposure. The insertion cost is located but not explained: a twenty word irrelevant sentence costs about two points and nothing in its content predicts how much. Within those bounds, verified exposure does not raise how often a specific answer comes back, and does raise how often the solution procedure comes back.

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A Related work

Huang et al. [Huang et al., 2025] construct hand authored hard perturbations of competition mathematics and attribute accuracy loss under an unchanged solution method to blind application of a learned procedure, a reading broadly consistent with the procedure level memorization in Section 3.2, established here against a corpus verified exactly rather than inferred. Wu et al. [Wu et al., 2025] show that reinforcement learning gains on mathematical reasoning depend on the pretraining exposure of the base policy and argue that clean synthetic problems are needed for reliable conclusions, a concern the checkpoint based design addresses by evaluating a policy before any relevant exposure occurred. Zhang et al. [Zhang et al., 2024] rebuild GSM8K from a contamination free procedure and report accuracy gaps of up to 13 points that track prior exposure across closed models, using an estimated exposure signal in the absence of any released training corpus; the present work substitutes an exact containment measurement for that estimate on a family where the corpus is public.

B Containment method, calibration, and corpus routes

The within split contamination contrast, high against low containment GSM8K train items, was closed rather than merely complicated by two findings. The corpus subset actually consumed during Instella 3B stage two pretraining, `train_119K`, is a 119,014 row subset of a 1,367,882 row released pool; containment computed against the pool over counts exposure, since only 7.56 percent of GSM8K train reaches 0.80 or higher against the subset actually consumed against 29.75 percent against the pool. Independently, `allenai/tulu-3-sft-mixture` is itself part of the Instella stage two pretraining mixture and `nvidia/OpenMathInstruct-2` enters at supervised fine tuning; both are generated in part from GSM8K train, and both carry it at containment 0.999 or higher on more than 97 percent of items. Every GSM8K train item therefore entered post stage one training through at least one of these routes regardless of its containment in the Instella specific synthetic corpus, so a high against low contrast defined on that corpus alone would compare two groups that are both almost entirely exposed once these mixtures are accounted for; the contrast is not attenuated by this, it is unidentifiable.

Item text is normalized to lowercase alphanumeric tokens and decomposed into every window of 13 consecutive tokens, following the overlap convention of Brown et al. [Brown et al., 2020]. Containment against a document is the fraction of an item’s windows found in that document; containment against a corpus is the maximum of that fraction over every document, computed in one streaming pass with an inverted index from window to source item so cost scales linearly in corpus size. Because the corpus is generated only from the GSM8K training split, every GSM8K test item is a verified negative, and the false positive rate is measured rather than assumed: at the adopted threshold of 0.80, zero of 1319 test items are flagged, a Wilson 95 percent upper bound of 0.29 percent; a looser rule flagging any window match flags 3 of 1000 sampled test items, 0.30 percent.

Within the Tulu 3 mixture, a sampled prefix of 300,000 rows, six of the mixture’s subsets, finds GSM8K train items reaching containment 0.999 through the `flan_v2_converted` subset, 6021 of 7473 items from 89,982 rows of that source, and through `wildchat` conversation logs, 4 items from 100,000 rows, an organic route in which users pasted benchmark problems into chat. Every scan reports rows examined alongside items found, since a sampled prefix bounds containment from below and a zero means absent from the rows scanned rather than absent from the corpus.

C Reproduction

All generation, containment, and analysis code, together with every scored generation file and containment record underlying this work, is released for audit at a public repository and a paired dataset mirror, both anonymized for review and to be de anonymized on acceptance. Every containment number reported uses the maximum over individual documents definition stated above; an earlier internal computation of the same quantity against the corpus pooled as one text produced a materially different figure for the same released split and is superseded by the document level definition throughout this work. Random seeds are fixed across item selection, deletion variant construction, the continued pretraining mixture, and every cluster bootstrap. An assistant was used to help implement generation, containment, and analysis code under the direction of the authors, who indepen-

259 dently verified every reported number against the underlying generation files; the assistant did not
260 design the probes or select which results to report.

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